

Number Theory: Divisibility and Modular Arithmetic

CS1010 Discrete Mathematics for Computer Science

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Why Study Number Theory?

Number theory is the study of the **integers**

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Although integers are familiar objects, they hide surprisingly rich and beautiful structures.

- ▶ When does one integer **divide** another?
- ▶ How do we find and work with **prime numbers**?
- ▶ How can we solve equations involving integers?
- ▶ What happens when we perform arithmetic with **remainders**?

And why does it matter?

These elementary ideas form the mathematical foundation of **modern cryptography** and have important applications throughout computer science.



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Divisibility and Modular Arithmetic

Division

Definition

Let a and b are integers and $a \neq 0$. We say **a divides b** if there exists an integer d such that:

$$b = ad \text{ and if } b \neq 0 \text{ then } |a| \leq |b|$$

- ▶ The notation $a \mid b$ denotes that a divides b .
- ▶ If a does not divide b , we write $a \nmid b$.

Terminology

Division Theorem

For any integer $a \neq 0$ and any positive integer b , there exists **unique** integers q and r such that:

$$b = aq + r \text{ and } 0 \leq r < b$$

- ▶ d is called the **divisor**.
- ▶ a is called the **dividend**.
- ▶ q is called the **quotient** ($q = a \text{ div } d$).
- ▶ r is called the **remainder** or the **residue** ($r = a \text{ mod } d$).

Q. What are the quotient and remainder when 87 is divided by 7?

A. The quotient is $12 = 87/7$, and the remainder is $3 = 87 \text{ mod } 7$.



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Properties of Divisibility

These properties allow us to manipulate divisibility statements algebraically, much like ordinary equations.

Theorem

Let a, b , and c be integers, where $a \neq 0$.

- (i) If $a \mid b$ and $a \mid c$, then $a \mid (\alpha b + \beta c)$ for any integers α and β
- (ii) If $a \mid b$, then $a \mid bc$ for all integers c
- (iii) If $a \mid b$ and $b \mid c$, then $a \mid c$

Proof: Exercise



Congruence Relation

Definition

Let a and b are integers and m is a positive integer. If $m|(a - b)$, then a is **congruent to b modulo m** , and we write $a \equiv b \pmod{m}$.

- ▶ m is called its **modulus**.
- ▶ If not congruent, we write $a \not\equiv b \pmod{m}$.
- ▶ $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$
 - ▷ i.e., they have the same remainder when divided by m .
- ▶ $a \equiv b \pmod{m}$ iff $a = b + km$ for some integer k .

Example: Determine if $21 \equiv 5 \pmod{4}$

- ▶ $21 \equiv 5 \pmod{4}$ because $4 \mid (21 - 5 = 16)$.



Congruences of Sums and Products

Can we perform arithmetic on congruences?

Theorem

Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

$$a + c \equiv b + d \pmod{m} \quad \text{and} \quad ac \equiv bd \pmod{m}$$

Example: Since $21 \equiv 1 \pmod{5}$ and $17 \equiv 2 \pmod{5}$:

- ▶ $38 = 21 + 17 \equiv 1 + 2 = 3 \pmod{5}$
- ▶ $357 = 21 \cdot 17 \equiv 1 \cdot 2 = 2 \pmod{5}$
- ▶ **Multiplication:** If $a \equiv b \pmod{m}$, then $c \cdot a \equiv c \cdot b \pmod{m}$.
- ▶ **Addition:** If $a \equiv b \pmod{m}$, then $c + a \equiv c + b \pmod{m}$.

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Arithmetic Modulo m

Definitions

Let $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$ be the set of nonnegative integers less than m :

- ▶ **Addition:** $a +_m b = (a + b) \bmod m$
- ▶ **Multiplication:** $a \cdot_m b = (a \cdot b) \bmod m$

▶ $12 +_{15} 11 = (12 + 11) \bmod 15 = 23 \bmod 15 = 8$

▶ $12 \cdot_5 11 = (12 \cdot 11) \bmod 5 = 132 \bmod 15 = 12$

Additive Inverses

If $a \neq 0$ belongs to $\mathbb{Z}_m$, then $m - a$ is the additive inverse of a modulo m , and 0 is its own inverse.

Arithmetic Modulo m : Algebraic Properties

The operations $+_m$ and $\cdot_m$ satisfy structural properties:

- **Closure:** If $a, b \in \mathbb{Z}_m$, then $a +_m b \in \mathbb{Z}_m$ and $a \cdot_m b \in \mathbb{Z}_m$.
- **Associativity:** $(a +_m b) +_m c = a +_m (b +_m c)$ and $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.
- **Commutativity:** $a +_m b = b +_m a$ and $a \cdot_m b = b \cdot_m a$.
- **Identity:** Elements 0 and 1 are the identities. $a +_m 0 = a$ and $a \cdot_m 1 = a$.
- **Distributivity:** $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$

Side Note

Using abstract algebra terminology, $\mathbb{Z}_m$ with $+_m$ is a **commutative group**, and $\mathbb{Z}_m$ with $+_m, \cdot_m$ is a **commutative ring**.

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Divisibility: Exercises

2. If $a, b, c, d \in \mathbb{Z}$, $a \neq 0$, and $a \mid c, b \mid d$, show that $ab \mid cd$.

Divisibility: Exercises

Q. If $a, b, c, d \in \mathbb{Z}$, $a \neq 0$, and $a \mid c, b \mid d$, show that $ab \mid cd$.

A. Since $c = ak$ and $d = b\ell$ for some $k, \ell \in \mathbb{Z}$, $cd = (ak)(b\ell) = ab(k\ell)$.

Q. If $a, b, c \in \mathbb{Z}$, $a \neq 0$, $c \neq 0$, and $ac \mid bc$, show that $a \mid b$.

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A. Since $bc = ack$ for some $k \in \mathbb{Z}$ and $c \neq 0$, $b = ak$.

Q. If a, b, c are positive integers and $a \mid bc$, must we have $a \mid b$ or $a \mid c$?

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Q. If a, b, c are positive integers and $a \mid bc$, must we have $a \mid b$ or $a \mid c$?

A. **No.** For example, $6 \mid (2 \cdot 3)$, $6 \nmid 2$, $6 \nmid 3$.

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A. If a is even, then $b = ak$ is even; hence, if b is odd, a must be odd.

Q. If $a, b \in \mathbb{Z}$ are nonzero, $a \mid b$, and $a + b$ is odd, show that a is odd.

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Q. If $a, b \in \mathbb{Z}$ are nonzero, $a \mid b$, and $a + b$ is odd, show that a is odd.

A. If a were even, then $b = ak$ would also be even, making $a + b$ even; hence a is odd.

Q. If $a \in \mathbb{Z}$ is not divisible by 3, show that $(a + 1)(a + 2)$ is divisible by 3.

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Q. If a is a positive integer, show that $4 \nmid a^2 + 2$.



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Q. If a is a positive integer, show that $4 \nmid a^2 + 2$.

A. If a is even, $a^2 + 2 \equiv 2 \pmod{4}$; if a is odd, $a^2 + 2 \equiv 3 \pmod{4}$. Thus $4 \nmid a^2 + 2$.

Congruence: Think of a Clock

A 12-hour clock “wraps around” every 12 hours.

Q. What time does a 12-hour clock read 80 hours after it reads 11:00?

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A. Since $80 \equiv 8 \pmod{12}$, this is the same as moving forward 8 hours:

$$11 + 8 = 19 \equiv 7 \pmod{12}.$$

7:00

Q. What time does it read 40 hours before 12:00?

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7:00

Q. What time does it read 40 hours before 12:00?

A. Since $40 \equiv 4 \pmod{12}$, move backward 4 hours:

$$12 - 4 = 8.$$

8:00

What Does the Clock Tell Us?

On a 12-hour clock, times that differ by a multiple of 12 represent the **same position on the clock**.

For example,

$$3, \quad 15, \quad 27, \quad 39, \dots$$

all correspond to the same position.

In other words,

$$3 \equiv 15 \equiv 27 \equiv 39 \pmod{12}.$$

We do not care about the exact number; we care about its **position modulo 12**. This idea of “wrapping around” is exactly what **congruence modulo m** captures.



Congruence: Exercises

2. Suppose $a \equiv 4 \pmod{13}$ and $b \equiv 9 \pmod{13}$. Find $c \in \mathbb{Z}_{13}$, such that $c \equiv a^2 + b^2 \pmod{13}$.

Congruence: Exercises

Q. Suppose $a \equiv 4 \pmod{13}$ and $b \equiv 9 \pmod{13}$. Find $c \in \mathbb{Z}_{13}$, such that $c \equiv a^2 + b^2 \pmod{13}$.

A. $a^2 + b^2 \equiv 4^2 + 9^2 = 97 \equiv 6 \pmod{13}$; hence $\boxed{c = 6}$.

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Q. Compute $((457 \bmod 23)(182 \bmod 23)) \bmod 23$.

A. $457 \equiv 20 \pmod{23}$ and $182 \equiv 21 \pmod{23}$, so $20 \cdot 21 = 420 \equiv 6 \pmod{23}$.

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Q. List all integers between -100 and 100 that are congruent to $-1 \pmod{25}$.

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A. The integers are $\boxed{-76, -51, -26, -1, 24, 49, 74, 99}$.



Thank You for your
attention!

Any questions?